



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to DURAI NAL FUEL

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 80.64 km
Estimated Duration: 1.9 hours
Adjusted Duration (Heavy Vehicle): 2.4 hours
Start: (11.4381, 77.8734)
End: (10.860567, 77.906068)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri, to VW64+FMR, Karur, encompasses key areas: SH 86 in Sullipalayam and Solasiramani - Pasur. This route is approximately 80.64 kilometers and is estimated to take about 1.91 hours for trucks carrying hazardous materials. The journey involves navigating through rural and semi-urban areas typical of Tamil Nadu roadways, along predominantly state highways and local roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. The region can expect hot and dry conditions from March to June, and heavy rainfall from October to December due to the northeast monsoon. Weather-related hazards include:

- **Monsoon Flooding:** Roads may become waterlogged, obscuring visibility and affecting road conditions.
- **Summer Heat:** Potential for overheating vehicles. Check coolant levels frequently.

3. Traffic Patterns and Congestion

There is moderate traffic on SH 86 with peak congestion during:

- **Mornings (8:00 AM - 10:00 AM):** School and commuter traffic.
- **Evenings (5:00 PM - 7:00 PM):** Return commuters and local market activities.

Sullipalayam and Solasiramani areas may experience localized congestion due to narrow roads and roadside markets.

4. Assessment of Road Quality and Infrastructure

The main route via SH 86 is reasonably well-maintained, but expect:

- **Potholes** and uneven surfaces, especially after monsoon.
- **Narrow Sections:** Some segments in rural areas can be narrow, requiring caution.
- **Limited Signage:** Be attentive to road markers and village entry points.

5. Alternative Routes for Emergencies

In case of emergencies, such as roadblocks or accidents:

- Route 1: Divert to NH 44, a major national highway, though it is longer.
- Route 2: Use SH 21 connecting Erode and Karur as an alternative.

6. Local Regulations on Hazardous Material Transport

- Strict adherence to vehicle permits for transporting hazardous materials.
- Compliance with local speed regulations.
- Restrictions on transporting hazardous materials during certain hours in densely populated areas.

7. Historical Incidents

There have been sporadic incidents involving heavy vehicles, typically focusing on:

- **Overtaken Trucks:** Due to sharp bends and narrow roads.

- **Spillages:** Lack of containment measures during accidents.

8. Environmental Considerations and Sensitive Areas

Focus on minimizing the impact on:

- **Agricultural Fields:** Risk of contamination from spills.
- **Water Bodies:** Bridges and crossings should be navigated with care to prevent pollution.

9. Communication Coverage

Relatively good mobile coverage along SH 86 with potential dead zones around:

- **Remote areas near Solasiramani.**
- **Isolated stretches with dense vegetation.**

10. Estimated Emergency Response Times

Emergency services are generally available but could be delayed in rural segments:

- **Urban Areas:** 15-30 minutes.
- **Rural Areas:** 45-60 minutes, depending on proximity to the nearest town.

12. Overall Summary of Risk Assessment

The route is generally manageable for transporting hazardous materials but poses risks such as:

- **Weather Vulnerability:** During monsoons.
- **Road Quality:** Impacting vehicle handling.
- **Traffic Management:** Due to narrow roads in village segments.
- **Limited Emergency Services Accessibility:** Impacts response time.
- **Compliance with Regional Regulations:** Necessary to avoid penalties and ensure safety.

Drivers should proceed with caution, particularly in bad weather and during village passages. Regular vehicle checks, preparation for unexpected weather, and adherence to route regulations will minimize many potential hazards.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	High	11.37862, 77.89488	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	High	11.30859, 77.92918	15 KM/Hr
7	Turn	Medium	11.24192, 77.87363	30 KM/Hr
8	Blind Spot	Blind Spot	11.24001, 77.87368	10 KM/Hr
9	Turn	High	11.24061, 77.86899	15 KM/Hr
10	Turn	Medium	11.24111, 77.86893	30 KM/Hr
11	Turn	Medium	11.24127, 77.86879	30 KM/Hr
12	Turn	High	11.24157, 77.86876	15 KM/Hr
13	Turn	High	11.24163, 77.86852	15 KM/Hr
14	Turn	High	11.24242, 77.86833	15 KM/Hr
15	Turn	Medium	11.24248, 77.86799	30 KM/Hr
16	Blind Spot	Blind Spot	11.24384, 77.86689	10 KM/Hr
17	Turn	High	11.24100, 77.86026	15 KM/Hr
18	Turn	Medium	11.24194, 77.85910	30 KM/Hr
19	Turn	Medium	11.24199, 77.85864	30 KM/Hr
20	Turn	High	11.22377, 77.83980	15 KM/Hr
21	Turn	Medium	11.21072, 77.83922	30 KM/Hr
22	Turn	Medium	11.12703, 77.85401	30 KM/Hr
23	Turn	Medium	11.05046, 77.92522	30 KM/Hr
24	Turn	Medium	10.95908, 77.91206	30 KM/Hr
25	Turn	High	10.95890, 77.91189	15 KM/Hr
26	Turn	High	10.95883, 77.91107	15 KM/Hr
27	Turn	Medium	10.95606, 77.91351	30 KM/Hr
28	Turn	Medium	10.95170, 77.91380	30 KM/Hr
29	Turn	Medium	10.91844, 77.91242	30 KM/Hr
30	Turn	Medium	10.91798, 77.91254	30 KM/Hr
31	Turn	High	10.91757, 77.91295	15 KM/Hr
32	Turn	Medium	10.91525, 77.91232	30 KM/Hr
33	Turn	Medium	10.90953, 77.91356	30 KM/Hr
34	Turn	Medium	10.90217, 77.91353	30 KM/Hr
35	Turn	High	10.89444, 77.91835	15 KM/Hr
36	Turn	High	10.89425, 77.91891	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
37	Turn	Medium	10.88533, 77.92242	30 KM/Hr
38	Turn	Medium	10.88353, 77.92517	30 KM/Hr
39	Turn	High	10.87500, 77.92565	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	clinic	Jai Sri Amman Clinic	11.1102489, 77.8579522	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.30859, 77.92918



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.24192, 77.87363



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.24001, 77.87368



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.24061, 77.86899



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.24111, 77.86893



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.24127, 77.86879



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.24157, 77.86876



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.24163, 77.86852



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.24242, 77.86833



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.24248, 77.86799



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.24384, 77.86689



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.24100, 77.86026



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.24194, 77.85910



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.24199, 77.85864



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.22377, 77.83980



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.21072, 77.83922



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.12703, 77.85401



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.05046, 77.92522



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.95908, 77.91206



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.95890, 77.91189



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.95883, 77.91107



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.95606, 77.91351



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.95170, 77.91380



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.91844, 77.91242



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.91798, 77.91254



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.91757, 77.91295



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.91525, 77.91232



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.90953, 77.91356



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.90217, 77.91353



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.89444, 77.91835



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.89425, 77.91891



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.88533, 77.92242



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.88353, 77.92517



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.87500, 77.92565

